

Exercise 1: SQL Database Setup & Data Insertion

Objective: Create a simple SQL database and insert data into two tables: customers and orders.

```
import sqlite3

# Create an in-memory SQLite database
conn = sqlite3.connect(':memory:')
cursor = conn.cursor()

# Create tables
cursor.execute("""CREATE TABLE customers (
    customer_id INTEGER PRIMARY KEY,
    name TEXT NOT NULL,
    country TEXT NOT NULL
)""")

cursor.execute("""CREATE TABLE orders (
    order_id INTEGER PRIMARY KEY,
    customer_id INTEGER,
    order_date TEXT NOT NULL,
    amount REAL,
    FOREIGN KEY (customer_id) REFERENCES customers (customer_id)
)""")

# Insert data into customers table
customers = [(1, 'Alice', 'USA'), (2, 'Bob', 'Canada'), (3, 'Charlie', 'UK')]
cursor.executemany("INSERT INTO customers VALUES (?, ?, ?)", customers)

# Insert data into orders table
orders = [(1, 1, '2024-09-01', 250.5), (2, 2, '2024-09-10', 150.0), (3, 1, '2024-09-15', 350.0)]
cursor.executemany("INSERT INTO orders VALUES (?, ?, ?, ?)", orders)

conn.commit()
print("Database setup complete!")
```

Exercise 2: SQL Query with Join

Objective: Retrieve data by joining customers and orders tables.

```
# SQL query to join customers and orders and retrieve information
query = """
    SELECT customers.name, customers.country, orders.order_id, orders.order_date,
    orders.amount
    FROM customers
    JOIN orders ON customers.customer_id = orders.customer_id
    """

# Execute the query
cursor.execute(query)
results = cursor.fetchall()

# Display results
for row in results:
    print(row)
```

Exercise 3: Exporting Data to CSV

Objective: Export the result of the SQL join query to a CSV file.

```
import csv

# Export results to a CSV file
with open('orders_report.csv', 'w', newline='') as file:
    writer = csv.writer(file)
    writer.writerow(['Customer Name', 'Country', 'Order ID', 'Order Date', 'Amount'])
    writer.writerows(results)

print("Data exported to 'orders_report.csv'!")
```

Exercise 4: Exporting Data to JSON

Objective: Export the same SQL join query result to a JSON file.

```
import json

# Convert the results into a list of dictionaries
data = []
for row in results:
    data.append({
        'Customer Name': row[0],
        'Country': row[1],
        'Order ID': row[2],
        'Order Date': row[3],
        'Amount': row[4]
    })

# Export the data to a JSON file
with open('orders_report.json', 'w') as file:
    json.dump(data, file, indent=4)

print("Data exported to 'orders_report.json'!")
```

Exercise 5: Importing Data from CSV

Objective: Load data from a CSV file into a database table.

```
# Create a new table for imported data
cursor.execute("""
    CREATE TABLE imported_orders (
        order_id INTEGER,
        customer_name TEXT,
        country TEXT,
        order_date TEXT,
        amount REAL
    )
""")

# Load data from CSV into the table
with open('orders_report.csv', newline='') as file:
    reader = csv.reader(file)
    next(reader) # Skip header row
    cursor.executemany('INSERT INTO imported_orders VALUES (?, ?, ?, ?, ?)', reader)

conn.commit()
```

```
# Verify that data is inserted correctly
cursor.execute('SELECT * FROM imported_orders')
print(cursor.fetchall())
```

Exercise 6: Importing Data from JSON

Objective: Load data from a JSON file into a database table.

```
# Create a new table for JSON-imported data
cursor.execute("""
    CREATE TABLE json_orders (
        customer_name TEXT,
        country TEXT,
        order_id INTEGER,
        order_date TEXT,
        amount REAL
    )
""")

# Load data from JSON into the table
with open('orders_report.json', 'r') as file:
    json_data = json.load(file)
    for entry in json_data:
        cursor.execute('INSERT INTO json_orders VALUES (?, ?, ?, ?, ?)',
            (entry['Customer Name'], entry['Country'], entry['Order ID'], entry['Order Date'], entry['Amount']))

conn.commit()

# Verify the data
cursor.execute('SELECT * FROM json_orders')
print(cursor.fetchall())
```